Israel-Stewart Causal Viscous Perturbations in the SSEE Dark Energy Framework: Exact Marginal Stability, ΛCDM-Consistent Structure Growth, and the Amplitude Ceiling

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Abstract

The Structural Self-Energy Expansion (SSEE) derives a dark energy equation of state $w_0 = -T_r/M_v \simeq -0.8399$ and $w_a \simeq -0.6700$ purely from the golden ratio φ and π, with zero fitted dimensionless parameters in the dark-energy background sector, in agreement with the official DESI DR2 w_0w_aCDM posterior at 0.24σ (Pantheon+; 0.2–1.8σ across SN compilations). (Following Paper 1 §1.3, the broader SSEE programme carries a small set of phenomenological ansätze in the dark-matter sector — in particular the ϕ-DM mass and the non-minimal coupling ξ — retracted with the φ-DM sector (Paper 6); OP-9/11 closed by dissolution, OP-14 resolved in `OPEN_PROBLEMS.md`; the present perturbation analysis is independent of those ansätze.) Because $w_0 < -1/3$, a naive k-essence interpretation gives a negative bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales.

We perform a linear Israel-Stewart (IS) causal viscous perturbation analysis of SSEE dark energy in the sub-horizon regime, addressing three interconnected questions: *(Q1)* Does IS regularise the gradient instability for all observationally relevant modes? *(Q2)* Do linear IS perturbations dynamically source the factor ≈ 2 that earlier drafts read between $\Omega_{m,\mathrm{CMB}}$ and the number $s_m = 0.160050$, or is there nothing to source (the former interpretation as a "MIRA factor" having been retired in the ω_m-direct reframe, OP-8 dissolved, and s_m being a quantity of the equation of state rather than a second matter sector)? *(Q3)* What are the falsifiable predictions for the growth rate, σ_8, and the S_8 parameter?

For Q1, we prove analytically that the SSEE structure constants satisfy the algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = |w_0|$, forcing $c^2_{s,\mathrm{eff}}(k \to \infty) = 0$ to floating-point precision (equivalently $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\mathrm{DE}}$, since $\Omega_{\mathrm{DE}} = |w_0|$ is the saturation correspondence of Paper 1, Postulate S, not an independent identity). This is not a numerical coincidence but a construction-level identity: any IS embedding of a dark-energy fluid satisfying $\tilde{\zeta} = |w_0|\,\tau_\Pi H_0$ is exactly marginally stable, and the SSEE structure

constants admit such an embedding. We do not claim this as an external theorem about all IS models, but as the consistency statement that SSEE supports a stable IS extension at the construction level. The critical wavenumber is $k_{\rm crit} = 0.456\, H_0/c$, placing the instability boundary *outside* the sub-horizon regime; all modes with $k > H_0/c$ are causally stabilised. The margin ($k_{\rm crit}/H_0 c^{-1} \simeq 0.46$, a factor ~ 2.2 inside the horizon) is comfortable at tree level. A future one-loop treatment would test whether the stability regime is robust to QFT corrections; we note this as an extension question rather than a current limitation.

For Q2, integrating the full linear IS perturbation ODE from $z = 19$ to $z = 0$ yields $\delta_{\rm DE}/\delta_m \to 0^-$ at all sub-horizon scales. Consequently the numerical enhancement ratio is $\Omega_{m,\rm eff}/\Omega_m = 0.9897 \pm 0.0167 \to 1$, not ≈ 2: linear IS perturbations generate no such link — and there is no second matter sector for them to link to. A background IS viscous calculation also fails to generate the factor (yielding ≈ 3.04, not ≈ 2). The CMB-to-dynamical ratio ≈ 1.93 is therefore not a fluid-dynamical effect. Nor is it a ratio of two matter densities, as earlier versions of this abstract stated: $\Omega_{m,\rm dyn} = 0.160050$ is not a density but $1 + w_0$, a quantity of the equation of state, so the difference $0.308881 - 0.160$ that once defined a second sector subtracts objects of different kinds. That construction is retracted (Paper 6). The matter content is a single sector, $\Omega_m = \omega_m/h^2 = 0.308881$, and the algebraic number MIRA $= (3\varphi + \pi)/4 \simeq 1.999$ survives only as the Hubble-screening amplitude of Paper 9, no longer as a matter factor.

For Q3, we compute the IS growth index $\gamma_{\rm IS} = 0.5504 \pm 0.0003$ (consistent with $\gamma_{\Lambda\rm CDM} \simeq 0.55$), the growth factor $\mathcal{G} = D_1^{\rm SSEE}/D_1^{\Lambda\rm CDM} = 1.0032$ (growth essentially identical to ΛCDM once the correct cosmological background $\Omega_{m,\rm cosm} = 0.308881$ is used in the Poisson source; see Paper 1, §"ω_m-direct" — the Two-Ω_m Criterion that once stated this rule is itself *withdrawn*, because $s_m = 0.160050$ was never a density that could compete for the slot), and the predicted $S_8 = \sigma_8(\Omega_{m,\rm CMB}/0.3)^{0.5}$. The amplitude is a separate matter and is not this paper's result. Fixing A_s to Planck gives $S_8^{\rm SSEE} = 0.8256 \pm 0.0061$, and the conservative CLASS ceiling is higher still, $S_8^{\rm ceiling} = 0.827$. Both are *ceilings under an imported normalisation*, not predictions: A_s is one of the two free CMB parameters of the model, and fixing it to Planck imports the Planck–KiDS offset instead of predicting it. With A_s free and this same background, the fit to the raw KiDS-1000 $\xi_\pm$ gives $S_8 = 0.7559 \pm 0.0189$, 0.10σ from KiDS-1000; against the later KiDS-Legacy release (357 raw $\xi_\pm$ points) the amplitude the shear asks for lies 0.49σ from the one the model's own CMB fit supplies, so A_s can be fixed there as well, and with *no* free cosmological parameter the model predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured (Paper 6).[1] What this paper establishes is the growth *shape*, $\mathcal{G}$ and $\gamma_{\rm IS}$, which are amplitude-independent. The $f\sigma_8$ mean tension is 0.70σ, essentially identical to ΛCDM (0.73σ): SSEE does not suppress or enhance structure formation relative to ΛCDM once $\Omega_{m,\rm cosm} = 0.308881$ governs the growth equation.

Finally, we map the SSEE IS parameters to the EFT-of-dark-energy operator basis [Gubitosi et al., 2013, Bellini and Sawicki, 2014], show that IS viscosity generates a unique combination of the $\bar{M}_2^4$ braiding operator and a non-minimal kinetic term, and demonstrate that the ghost condensate [Arkani-Hamed et al., 2004] and

[1] KiDS-Legacy (Wright et al. 2025, arXiv:2503.19441) was published sixteen months before this analysis; no temporal priority is claimed. Its recalibration of the source redshift distributions dissolves the Planck–shear disagreement in the data themselves: re-running the KiDS-1000 analysis under blinding, the collaboration's Appendix I finds $2.39\sigma \to 1.42\sigma$ from the $n(z)$ calibration alone. The three χ^2 compared span 1.0 over 357 points, so the shear does *not* discriminate between the models; what separates them is the parameter count. See Paper 6.

SSEE IS mechanisms are physically inequivalent: SSEE achieves *exact* marginal stability ($c_s^2 = 0$, the minimal-viscosity solution), while ghost condensation gives $c_s^2 = 1$.

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1 Introduction

1.1 The SSEE framework and the perturbation challenge

The Structural Self-Energy Expansion (SSEE) [Almeida Vallejo, 2026a] is a minimal-parameter dark energy framework (zero fitted dimensionless parameters in the background sector; see Paper 1 §1.3) whose equation-of-state parameters

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi+\beta)}{\varphi+\pi+K_v} \simeq -0.8399, \qquad w_a = -\frac{P_{sc}}{I_g} \simeq -0.66997, \tag{1}$$

are derived algebraically from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π, without any continuous free parameters. Here $\beta = (\pi+\varphi)/2$, $K_v = \varphi+\pi+(\pi+\varphi)$, $T_r = 3(\varphi+\beta)$, $M_v = \varphi+\pi+K_v$, and $P_{sc} = (\pi+\varphi)+\varphi$. The CPL parameters (1) [Chevallier and Polarski, 2001, Linder, 2003] are consistent with DESI DR2 at 0.24σ (Pantheon+) [Almeida Vallejo, 2026c, Abdul-Karim et al., 2025] and with Planck 2018 CMB spectra at $\chi^2_r \leq 1.062$ [Almeida Vallejo, 2026d].

Paper 4 [Almeida Vallejo, 2026e] identifies the algebraic structure constants

$$\mathrm{KAL}_0 \equiv \beta+\pi \simeq 5.5214 \quad \text{(Structural Retention)}, \tag{2}$$

$$\mathrm{MIRA} \equiv \frac{3\varphi+\pi}{4} \simeq 1.9989 \quad \text{(MIRA factor)}, \tag{3}$$

and (under the ω_m-direct reframe; OP-8 dissolved) the CMB matter density is the standard physical $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 \simeq 0.308881$, built from the algebraic $\omega_m = \omega_b+\omega_c+\omega_\nu$, in 0.88σ agreement with Planck 2018. (The earlier MIRA mapping $\Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} \simeq 0.3199$ is superseded.)

The present paper confronts a fundamental question that must be answered before SSEE can claim theoretical consistency: *does the dark energy sector remain perturbatively stable*?

1.2 The gradient instability problem

A canonical-kinetic (constant-w) k-essence proxy with Lagrangian $\mathcal{L} \propto X^n$ [Armendariz-Picon et al., 2001], where $X = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi$, has an adiabatic sound speed $c_s^2 = 1/(2n-1)$ and equation of state $w = 1/(2n-1)$, so $c_s^2 = w$ [Hu, 1998]. For SSEE, $w_0 \simeq -0.840$, giving

$$c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840. \tag{4}$$

Negative c_s^2 signals a *gradient instability*: perturbations $\delta_{\mathrm{DE}} \propto e^{|c_s|kt}$ grow exponentially on a timescale $(|c_s|k)^{-1}$, which for $k \gtrsim 10\,H_0/c$ (BAO scales) is shorter than a Hubble time [Caldwell, 2002]. Left unchecked, this instability would generate excessive DE density contrast $|\delta_{\mathrm{DE}}| \gg 1$, invalidating linear perturbation theory and producing observable signatures in the CMB lensing and matter power spectra inconsistent with data [Calabrese et al., 2011].

We stress that this bare instability is an artefact of the *effective-fluid* (constant-w) closure $c_s^2 = w$, *not* a property of the fundamental SSEE field. The dark-energy action is the ghost condensate of Paper 7, $K = c_1X + c_2X^2$ with $c_2X/c_1 = -0.522735$, which is independently ghost-free and gradient-stable with adiabatic sound speed

$$c^2_{s,\mathrm{ad}} = \frac{1+2u}{1+6u} = \frac{M_v - T_r}{5M_v + 3T_r} = 0.021284, \qquad u \equiv c_2X/c_1, \tag{5}$$

a clustered rather than smooth dark energy. *Correction (2026-09-06).* Earlier versions of this paragraph quoted $c^2_{s,\mathrm{ad}} \in [0.60, 1]$ and attributed the Lagrangian to "Papers 7 and 10". That value was computed from $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$, which is Paper 10's *screening* functional and not the dark-energy action: the two carry different coefficient ratios ($+0.008577$ against -0.522735) and do different jobs [Almeida Vallejo, 2026b]. The sound speed of the dark-energy sector is the one above. The IS analysis below addresses the *hydrodynamic* (fluid) description used by Boltzmann codes, where dark energy is parametrised by (w, c_s^2) and the naive closure $c_s^2 = w$ would be catastrophic. The IS bulk viscosity is $\tilde{\zeta} = \mathrm{KAL}_0/3$, so the fluid-level regularisation introduces no new parameter.

What is and is not fixed here (2026-09-07). Only the *ratio* enters the observable: $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\mathrm{DE}}$, and since $\tilde{\zeta}$ and τ_Π share the factor KAL_0, it cancels. The marginal-stability result $c^2_{s,\mathrm{eff}} = w_0 + \Omega_{\mathrm{DE}} = 0$ is therefore *independent of* KAL_0: replacing it by any positive constant returns the same 0, which follows from the algebraic identity $\Omega_{\mathrm{DE}} = |w_0|$ alone. The individual split $\tilde{\zeta} = \mathrm{KAL}_0/3$ is consequently not determined by anything in *this* observable, and is not claimed to be.

It does not follow that the IS constants are unconstrained. KAL_0 enters the framework in three places that do move observables: $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ (a 10% shift moves ω_c by $\sim 10\sigma$ against Planck); the neutrino sum $\Sigma m_\nu = R_2\, \omega_b\, (93.14)/(\tau_\Pi H_0)$, where KAL_0 appears *squared* through $R_2 = \Omega/(\mathrm{KAL}_0 T_r)$ and $\tau_\Pi H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}})$, so that $\mathrm{KAL}_0 \to 1.1\,\mathrm{KAL}_0$ gives $\Sigma m_\nu = 0.0566$ eV, *below* the oscillation floor 0.058 eV and therefore falsified; and Paper 10's screening functional. The cancellation above is thus local to $c^2_{s,\mathrm{eff}}$: τ_Π itself is anchored, through Σm_ν, to within roughly 10%.

Correction (2026-09-07). Earlier versions of this passage read "the same structural constant KAL_0 that normalises $K(X)$ also fixes the IS bulk viscosity". That is wrong about *which* $K(X)$: KAL_0 does not appear in the dark-energy action at all — Paper 7's kinetic function is $K = c_1 X + c_2 X^2$, and the X/KAL_0 form belongs to Paper 10's screening functional. Worse, the dark-energy action *cannot* supply a bulk viscosity: for shift-symmetric $K(X)$ the pressure perturbation is exactly adiabatic, $\delta p - c_s^2\, \delta\rho = 0$ identically in (c_1, c_2, X), so the entropy production of Eq. (6) vanishes and $\zeta = 0$. The IS layer is therefore a repair of the (w, c_s^2) *parametrisation* imposed by Boltzmann codes, not a property of the field; the sector's own prediction is the field value $c_s^2 = 0.021284$. The recurrence of KAL_0 in $\tilde{\zeta}$ is a resemblance across two different objects, not a derivation of one from the other, and is not claimed as one. See **OP-22b**.

1.3 The Israel-Stewart resolution

The first-order (Eckart) relativistic fluid theory [Eckart, 1940] is acausal [Müller, 1967] and generically unstable [Hiscock and Lindblom, 1985]: perturbations propagate at infinite speed. Israel [Israel, 1976] and Israel & Stewart [Israel and Stewart, 1979] constructed a second-order causal theory in which dissipative fluxes satisfy relaxation equations with a finite relaxation time τ_Π. Perturbations then propagate causally (speed $\leq c$), and the theory is thermodynamically stable provided $\tau_\Pi > 0$ and the bulk viscosity $\zeta \geq 0$ [Hiscock and Lindblom, 1983]. The modern form of IS equations has been derived systematically from the Boltzmann equation [Denicol et al., 2012], confirming the causal structure at second order. Bulk-viscous cosmological models have been explored phenomenologically in Brevik and Grøn [2013], motivating the present IS dark-energy application.

SSEE identifies in Paper 1 a structural IS steady-state condition that fixes KAL_0 as the dimensionless viscosity, implying specific values of both ζ and τ_Π with no additional

freedom. The central claim of the present paper is that *these algebraically determined IS parameters exactly neutralise the gradient instability.*

1.4 Paper structure

Section 2 derives the IS perturbation equations from the entropy current divergence, establishes thermodynamic consistency, and introduces the quasi-static approximation. Section 3 establishes the marginal stability identity (under minimality) and characterises the critical wavenumber. Section 4 presents numerical integration results for the MIRA test. Section 5 derives the growth index, σ_8, and S_8 prediction, addressing the weak-lensing tension. Section 6 maps the IS mechanism to the EFT-of-dark-energy operator basis. Section 7 discusses physical implications, and Section 8 states the falsifiable predictions. Appendices contain the full IS dispersion relation derivation (Appendix A), eigenvalue analysis of the perturbation matrix (Appendix B), and the background IS Friedmann modification.

2 Israel-Stewart Theory for SSEE Dark Energy

2.1 Entropy production and the second-order IS equations

In relativistic thermodynamics the second law requires $\nabla_\mu S^\mu \geq 0$, where $S^\mu = su^\mu - q^\mu/T + \cdots$ is the entropy 4-current [Weinberg, 1971, Israel, 1976, Israel and Stewart, 1979]. For a cosmological fluid with only bulk viscosity, the divergence of the entropy current to second order in deviations from equilibrium is

$$\nabla_\mu S^\mu = \frac{1}{T}\left[-\Pi\,\theta - \frac{\tau_\Pi}{2\zeta T}\,\Pi\,\dot{\Pi} - \frac{\Pi^2}{\zeta}\right] \geq 0, \tag{6}$$

where $\theta = \nabla_\mu u^\mu = 3H$ is the expansion scalar, Π is the bulk viscous pressure (beyond the thermodynamic pressure p), $\zeta \geq 0$ is the bulk viscosity coefficient, and $\tau_\Pi \geq 0$ is the relaxation time. Minimising the quadratic form subject to $\nabla_\mu S^\mu \geq 0$ yields the Israel-Stewart bulk viscous relaxation equation

$$\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H - \frac{\tau_\Pi\,\Pi}{2}\left[\frac{\dot{\tau_\Pi}}{\tau_\Pi} - \frac{\dot{\zeta}}{\zeta} - \frac{\dot{T}}{T}\right], \tag{7}$$

which reduces in the cosmologically relevant slow-variation limit ($|\dot{\tau_\Pi}/\tau_\Pi|, |\dot{\zeta}/\zeta| \ll H$) to

$$\boxed{\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H.} \tag{8}$$

This is the form used throughout this paper. Two properties follow immediately from Eq. (6):

1. **Causality**: the IS relaxation limits signal propagation to $v_{\rm sig} = \sqrt{\zeta/[\tau_\Pi(\rho_{\rm DE} + p_{\rm DE})]} < c$ [Hiscock and Lindblom, 1983] — the enthalpy again, as the inertia of the propagating mode.

2. **Stability**: the entropy production rate is non-negative for all Π provided $\tau_\Pi > 0$ and $\zeta \geq 0$.

Both conditions are satisfied in SSEE (established in Section 3).

2.2 SSEE IS parameters from algebraic structure constants

In the SSEE steady state $\dot{\Pi} = 0$, Eq. (8) gives $\Pi = -3\zeta H$. The dimensionless viscosity and relaxation time are defined by normalising to the dark-energy *enthalpy* and the Hubble rate:

$$\tilde{\zeta} \equiv \frac{\zeta}{(\rho_{\rm DE} + p_{\rm DE})\, H_0} = \frac{\zeta}{\rho_{\rm DE}(1 + w_0)\, H_0}, \tag{9}$$

$$\tau_\Pi H_0 \equiv \text{dimensionless IS relaxation time.} \tag{10}$$

The enthalpy $\rho + p$, not the density ρ, is the correct normalisation: it is the inertia that appears in the relativistic Euler equation (Appendix A, second line) and the origin of the $1/(1 + w)$ factor multiplying the pressure gradient in the standard fluid perturbation equations. Normalising this way makes $\tilde{\zeta}/(\tau_\Pi H_0)$ directly a squared sound speed, with no residual factors.

Resolved: the normalisation of $\tilde{\zeta}$ (OP-22, closed 2026-09-06)

A previous version of this box reported $c^2_{s,\mathrm{eff}} = 0$ as an artefact of a normalisation mismatch between documents. That diagnosis was wrong, and we correct it here rather than leave it standing.

The question. $\tilde{\zeta} = \mathrm{KAL}_0/3$ is a dimensionless viscosity, but dimensionless *with respect to what*? The candidates differ by large factors, because one physical quantity $\zeta/(\rho\,\tau_\Pi)$ takes three values depending on the divisor:

divisor	$\zeta/(\rho\,\tau_\Pi)$
ρ_{crit} (total)	0.691119
ρ_{DE}	1.000000
$\rho_{\mathrm{DE}} + p_{\mathrm{DE}}$ (enthalpy)	6.248039

The evidence, and it is not unanimous. Three places in the framework fix the normalisation, and one of them points the other way. The steady-state ansatz as originally written, $\Pi = -\mathrm{KAL}_0\,\rho_{\mathrm{DE}} H$, gives $\zeta = \mathrm{KAL}_0\rho_{\mathrm{DE}}/3$ and hence $c^2_{s,\mathrm{eff}} = w_0 + 5.248 = +4.41$: superluminal, and therefore not an option. The two operative appendices of this paper both require the other convention.

The answer, from two independent witnesses inside this paper. The relativistic inertia of a sound wave is the enthalpy, so the dispersion relation of Appendix A carries $\zeta/[\tau_\Pi(\bar{\rho}+\bar{p})]$; it reduces to $\tilde{\zeta}/(\tau_\Pi H_0)$ *only if* $\tilde{\zeta}$ is enthalpy-normalised, $\zeta = \tilde{\zeta}\,(\rho_{\mathrm{DE}}+p_{\mathrm{DE}})H_0$, which is what that appendix states. Independently, the eigenvalue analysis of Appendix B uses $\mathcal{F} = (1-3c_s^2)+\tilde{\zeta}(k/H)^2 \approx 186$ at $k = 10$: the enthalpy normalisation returns 185.05, whereas normalising to ρ_{crit} returns 1370, an order of magnitude off. Both appendices therefore use the same convention, and it is the enthalpy one.

The resolution, and why it is not merely a choice. We adopt the enthalpy form of the steady-state condition, $\Pi = -\mathrm{KAL}_0\,(\bar{\rho}+\bar{p})\,H$. The decisive argument is a limit test, independent of anything in SSEE. Take $w \to -1$ exactly. Then $\bar{\rho}+\bar{p} \to 0$: a cosmological constant carries no fluid degrees of freedom, so its bulk viscous pressure *must* vanish. The enthalpy form gives $\Pi \to 0$ automatically; the ρ_{DE} form gives a finite viscous pressure for a cosmological constant, which is not a viable hydrodynamic limit. Two further reasons agree: it is what both appendices of this paper already use, and it is the form consistent with relativistic dissipative hydrodynamics, where the enthalpy is the inertia in the Euler equation and in the bulk-viscous transport alike [Maartens, 1996, Hiscock and Lindblom, 1985]. That the alternative is superluminal is then a consequence, not the reason for choosing. The archived note in which the ρ_{DE} form was first written is superseded by this.

Consequence. $c^2_{s,\mathrm{eff}} = w_0 + \tilde{\zeta}/(\tau_\Pi H_0) = w_0 + \Omega_{\mathrm{DE}} = 0$ stands: marginally stable, manifestly subluminal, and **not** an artefact. The 0 is a result.

What is *withdrawn.* The *justification* of τ_Π given in the EFT appendix of Paper 1 [Almeida Vallejo, 2026a], which claimed to derive $\tau_\Pi H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}})$ by saturating the causality bound written as $\zeta/(\rho_{\mathrm{DE}}\tau_\Pi) \le 1$. That bound uses ρ where the enthalpy belongs; imposed correctly it returns $\tau_\Pi H_0 = \tilde{\zeta}(1+w_0) = 0.2946$, not 2.191. The factor between them is exactly $(1+w_0)^{-1} = 6.248$. Nothing numerical moves — $\tau_\Pi H_0 = \mathrm{KAL}_0\Omega/T_r$ is algebra, and was already stated in that form in Eq. (11) — but causality does not *derive* it, and Paper 1 no longer claims that it does.

Residual, still open — and smaller than it looked. The framework carries two sound speeds for the dark-energy sector:

mode	source	c_s^2

With the steady-state condition as currently written, and combined with the IS equation, one obtains [Almeida Vallejo, 2026a]:

$$\boxed{\tilde{\zeta} = \frac{\mathrm{KAL}_0\,\Omega}{M_v} = \frac{\mathrm{KAL}_0}{3} \simeq 1.8405, \qquad \tau_\Pi H_0 = \frac{\mathrm{KAL}_0\,\Omega}{T_r} \simeq 2.191.} \tag{11}$$

Written this way the two are manifestly the *same* construction, built from the same parents (KAL_0, Ω) and differing only in which sovereignty sits in the denominator: T_r (TRIAL) for the relaxation time, M_v (ATLAS) for the viscosity. The factor 3 in $\mathrm{KAL}_0/3$, which looks arbitrary in isolation, is simply $M_v = 3\Omega$. Both parameters are determined solely by $\mathrm{KAL}_0 = \beta + \pi$, $\Omega = \varphi + \pi$ and the sovereignties — no freedom remains. The ratio

$$\frac{\tilde{\zeta}}{\tau_\Pi H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3\Omega_{\mathrm{DE}})} = \Omega_{\mathrm{DE}} = |w_0| \tag{12}$$

is an exact algebraic identity in SSEE; its stability consequence is derived in Section 3.

2.3 Thermodynamic consistency

The IS theory is thermodynamically consistent if and only if:

(i) $\tau_\Pi > 0$: satisfied, $\tau_\Pi H_0 = 2.191 > 0$.

(ii) $\zeta \geq 0$: satisfied, $\tilde{\zeta} = 1.841 > 0$.

(iii) The signal speed $v_{\mathrm{sig}} = \sqrt{\zeta/(\tau_\Pi \rho_{\mathrm{DE}})} = \sqrt{\tilde{\zeta}/(\tau_\Pi H_0)/\Omega_{\mathrm{DE}}}$: in SSEE: $v_{\mathrm{sig}}/c = \sqrt{|w_0|/\Omega_{\mathrm{DE}}} = \sqrt{1} = 1$, so $v_{\mathrm{sig}} = c$ exactly. The signal speed saturates the causality bound.

The fact that $v_{\mathrm{sig}} = c$ is another algebraic consequence of Eq. (12): $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\mathrm{DE}} = |w_0|$, and $v_{\mathrm{sig}}^2/c^2 = \tilde{\zeta}/((\tau_\Pi H_0)\Omega_{\mathrm{DE}}) = 1$. This means SSEE dark energy perturbations propagate exactly at the speed of light in the IS high-k limit — the maximally causal configuration consistent with thermodynamic stability.

2.4 Linearised perturbation equations in Newtonian gauge

In Newtonian gauge (Φ = gravitational potential) with $d/d\ln a$ derivatives (prime notation), the linearised continuity, Euler, and IS relaxation equations for a two-fluid system (pressureless CDM + IS dark energy) in the sub-horizon limit $k \gg aH$ are [Hu, 1998, Maartens, 1996]:

$$\delta_m' = -\frac{k}{aH}\, v_m + 3\Phi', \tag{13}$$

$$v_m' = -v_m - \frac{k}{aH}\,\Phi, \tag{14}$$

$$\delta_{\mathrm{DE}}' = -(1+w)\left[\frac{k}{aH}\, v_{\mathrm{DE}} - 3\Phi'\right], \tag{15}$$

$$v_{\mathrm{DE}}' = -(1-3c_s^2)\, v_{\mathrm{DE}} - \frac{k}{aH}\left[\frac{c_s^2\,\delta_{\mathrm{DE}}}{1+w} + \frac{\Phi}{1+w} - \tilde{\Pi}\right], \tag{16}$$

$$\tau_\Pi H\, \tilde{\Pi}' = -\tilde{\Pi} - \tilde{\zeta}\,\frac{k}{aH}\, v_{\mathrm{DE}}, \tag{17}$$

where $w = w_0 + w_a(1-a)$ is the CPL equation of state, $\tilde{\Pi}$ is the dimensionless bulk viscous pressure perturbation normalised to $\rho_{\rm DE}$, and the Newtonian potential satisfies

$$\frac{k^2}{a^2}\Phi = -\frac{3H_0^2}{2a}\left[\Omega_m(a)\,\delta_m + \Omega_{\rm DE}\,f_{\rm DE}(a)\,\delta_{\rm DE}\right], \tag{18}$$

with $f_{\rm DE}(a) = \rho_{\rm DE}(a)/\rho_{\rm DE,0}$. The $3\Phi'$ terms in Eqs. (13)–(15) arise from the metric perturbation and are retained for completeness; in the deep sub-horizon limit $k \gg aH$ they are subdominant and we drop them in the numerical integration, consistent with standard practice [Hu, 1998].

2.5 Quasi-static IS approximation and stiffness reduction

The IS relaxation eigenvalue is $\lambda_\Pi \approx -(\tau_\Pi H)^{-1}$. At $z = 0$, $\tau_\Pi H = \tau_\Pi H_0 \cdot (H/H_0) = 2.191$; at $z = 99$, $H/H_0 \approx 556$, so $\tau_\Pi H \approx 1218$ (Table 1). The Eq. (17) ODE is therefore stiff: the $\tilde{\Pi}$ mode relaxes on a timescale $\sim 1/(1218\,H)$ at early times, orders of magnitude faster than the growing mode.

Setting $\tilde{\Pi}' = 0$ in Eq. (17) (quasi-static IS approximation, valid when $\tau_\Pi H \gg 1$) yields the algebraic relation

$$\tilde{\Pi}_{\rm QS} = -\tilde{\zeta}\,\frac{k}{aH}\,v_{\rm DE}, \tag{19}$$

which substituted into Eq. (16) gives an effective velocity friction:

$$v'_{\rm DE} = -\underbrace{\left[(1-3c_s^2) + \tilde{\zeta}\left(\frac{k}{aH}\right)^2\right]}_{\mathcal{F}(k,a)} v_{\rm DE} - \frac{k}{aH}\,\frac{c_s^2\,\delta_{\rm DE}}{1+w} - \frac{k}{aH}\,\frac{\Phi}{1+w}. \tag{20}$$

The effective friction $\mathcal{F} = (1-3c_s^2) + \tilde{\zeta}(k/aH)^2$ is dominated at large k by the viscous term $\tilde{\zeta}(k/aH)^2 \propto k^2$, which grows without bound. This k^2 suppression is the physical mechanism behind IS stabilisation of DE perturbations. The 5-variable stiff ODE system (Eqs. 13–17) is thus reduced to a 4-variable system $(\delta_m, v_m, \delta_{\rm DE}, v_{\rm DE})$ integrable with standard Runge-Kutta methods.

Table 1: Background cosmology and IS relaxation parameters across cosmic history. H/H_0 is computed from the SSEE CPL Friedmann equation with the matter density $\Omega_m = 0.308881$ and CPL dark energy.$\tau_\Pi H$ shows that the quasi-static approximation ($\tau_\Pi H \gg 1$) is excellent at all redshifts.

z	a	H/H_0	$\tau_\Pi H$	QS validity
99	0.010	555.8	1218	$\gg 1$
19	0.050	49.71	108.9	excellent
9	0.100	17.58	38.52	excellent
4	0.200	6.237	13.67	very good
1	0.500	1.768	3.873	adequate
0.25	0.800	1.157	2.534	marginal
0	1.000	1.000	2.191	marginal

QS error estimate. At $z = 0$, $\tau_\Pi H = 2.191$, so the adiabatic correction to the quasi-static approximation is of order $(\tau_\Pi H)^{-1} \simeq 0.46$. However, this error in Π propagates to matter growth through a two-step coupling: $\Delta\Pi \to \Delta v_{\rm DE} \to \Delta\delta_{\rm DE} \to \Delta\Phi \to \Delta\delta_m$. At sub-Hubble scales, the IS effective friction in the DE Euler equation is $\mathcal{F} \approx \tilde{\zeta}(k/aH)^2$; for $k \geq 5\,H_0/c$ this gives $\mathcal{F} \geq 46$, so the fractional error in $v_{\rm DE}$ is $\lesssim 0.46/46 \approx 1\%$. The resulting error in the growth factor D_1 is further suppressed by $\Omega_{\rm DE}\,\delta_{\rm DE}/(\Omega_m\,\delta_m) \lesssim 5\%$ at $k \geq 5\,H_0/c$, giving $\Delta D_1/D_1 \lesssim 0.05\%$ for the BAO scales that dominate the σ_8 integral. Only super-Hubble modes ($k \sim H_0/c$) carry an $\mathcal{O}(25\%)$ DE-sector error, but these modes contribute negligibly to σ_8 (filtered by $W(8\,{\rm Mpc}/h)$). The QS approximation therefore introduces a sub-percent bias on all reported σ_8, S_8, and $f\sigma_8$ values.

3 Q1: Gradient Stability — Exact Marginal Stability Theorem

3.1 IS dispersion relation from linear perturbation theory

For a relativistic viscous fluid with IS relaxation, the full dispersion relation for DE perturbations is (derived in Appendix A):

$$\omega^2 = c_{s,\rm bare}^2\,k^2 + i\omega\,\frac{\zeta\,k^2}{\rho_{\rm DE}\,(\tau_\Pi\omega + i)} - \frac{k^2}{a^2}\,\frac{3H^2\Omega_{\rm DE}}{k^2/a^2}. \tag{21}$$

In the short-wavelength limit $k\tau_\Pi \gg aH$ (IS regime, $|\omega|\tau_\Pi \gg 1$), the damping term saturates and the dispersion relation simplifies to

$$\omega^2 = \left(c_{s,\rm bare}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\right) k^2 \equiv c_{s,\rm eff}^2\,k^2. \tag{22}$$

The full k-dependent interpolation between the Navier-Stokes ($k \to 0$) and IS ($k \to \infty$) limits is

$$c_{s,\rm eff}^2(k) = c_{s,\rm bare}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\cdot\frac{x^2}{1+x^2}, \quad x \equiv \frac{k\,c\,\tau_\Pi H_0}{aH}, \tag{23}$$

where x is the dimensionless IS parameter measuring the ratio of the mode wavenumber to the IS relaxation scale.

3.2 Marginal stability under minimality

SSEE Marginal Stability (given minimality)

In the SSEE dark energy framework, the IS parameters satisfy

$$\frac{\tilde{\zeta}}{\tau_\Pi H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3\,\Omega_{\mathrm{DE}})} = \Omega_{\mathrm{DE}} = |w_0|, \tag{24}$$

and therefore

$$c^2_{s,\mathrm{eff}}(k \to \infty) = c^2_{s,\mathrm{bare}} + |w_0| = w_0 + |w_0| = 0. \tag{25}$$

This is an exact algebraic identity within the SSEE constants, not a numerical coincidence. It expresses a construction-level consistency rather than a theorem constraining all IS models: SSEE supports an IS embedding for which dark-energy perturbations are exactly marginally stable in the high-k limit.

The proof follows directly from Eq. (12). The numerical verification gives $|c^2_{s,\mathrm{eff}}| = 3.1 \times 10^{-16}$ (floating-point round-off, confirming machine-precision exactness).

The physical significance is the following:

- The gradient instability present for $k < k_{\mathrm{crit}}$ (Navier-Stokes regime) is completely neutralised for $k > k_{\mathrm{crit}}$ (IS regime).
- The neutralisation is *exact*: $c^2_{s,\mathrm{eff}} = 0$ means DE perturbations propagate as effectively pressureless dust at small scales, exactly as required for consistency with CMB and LSS observations.
- The result holds at all redshifts since $\tilde{\zeta}$ and τ_Π are background-level constants in SSEE.

3.3 Uniqueness of the minimal-viscosity solution

The condition $c^2_{s,\mathrm{eff}} = 0$ requires $\tilde{\zeta}/(\tau_\Pi H_0) = |c^2_{s,\mathrm{bare}}|$. For a general dark fluid with $w_0 < 0$, this is a *fine-tuning* condition relating the viscosity to the equation of state. In SSEE both sides equal $\Omega_{\mathrm{DE}} = |w_0|$, and the KAL_0 dependence cancels in the ratio $\tilde{\zeta}/(\tau_\Pi H_0) = (\mathrm{KAL}_0\Omega/M_v)/(\mathrm{KAL}_0\Omega/T_r) = T_r/M_v$. No number is tuned.

We are explicit about what does the work, because it is easy to overclaim here. The relaxation time $\tau_\Pi H_0 = \mathrm{KAL}_0\Omega/T_r$ is fixed independently in Paper 1 [Almeida Vallejo, 2026a], and $w_0 = -T_r/M_v$ is fixed independently in the background sector. Given those two, requiring $c^2_{s,\mathrm{eff}} = 0$ leaves *no* freedom in $\tilde{\zeta}$: it must equal $|w_0|\,\tau_\Pi H_0 = \mathrm{KAL}_0\Omega/M_v$, with the T_r cancelling identically. So $\tilde{\zeta}$ is not an independent input.

What remains an input is the *requirement* of marginal stability itself, and it is a selection principle rather than a fitted number. Its justification is minimality: since $c^2_{s,\mathrm{bare}} = w_0 < 0$ is a gradient instability, some viscosity is mandatory, and $\tilde{\zeta} = \mathrm{KAL}_0\Omega/M_v$ is the *smallest* value that removes it — any less leaves the instability, any more is unmotivated surplus. We therefore state the result as a consequence of minimality, not as a prediction free of assumptions.

3.4 Critical wavenumber and observational window

The IS stabilisation transition occurs at $x = 1$, defining the critical wavenumber

$$k_{\rm crit}(a) = \frac{aH}{c\,\tau_\Pi H_0}. \tag{26}$$

At $z = 0$ ($a = 1$, $H = H_0$):

$$\frac{k_{\rm crit}}{H_0/c} = \frac{1}{\tau_\Pi H_0} = \frac{1}{2.191} \simeq 0.456. \tag{27}$$

Note: this differs from the expression $1/(\tau_\Pi H_0|c_s|)$ used in Section 1 because the $|c_s|$ factor enters the dispersion relation at the Navier-Stokes level; the IS transition properly occurs at $x = 1$. Since $k_{\rm crit} \simeq 0.456\,H_0/c < H_0/c$, the critical scale lies *below* the Hubble wavenumber. Every sub-horizon mode satisfies $k > aH/c = H_0/c$ at $z = 0$, i.e. $k > k_{\rm crit}$: all sub-horizon modes are IS-stabilised.

At higher redshift $k_{\rm crit}(z) = k_{\rm crit}(0) \times H(z)/H_0$, which grows with redshift. However, for a given comoving mode k, it entered the IS-stable regime at $a_{\rm stab}(k) = k\,c\,\tau_\Pi H_0/H(a_{\rm stab})$. For BAO scales ($k \approx 0.1\,{\rm Mpc}^{-1} \approx 230\,H_0/c$), $a_{\rm stab} \ll 10^{-3}$: these modes were IS-stable well before matter-radiation equality. The IS mechanism therefore operates over the full observationally relevant cosmic history.

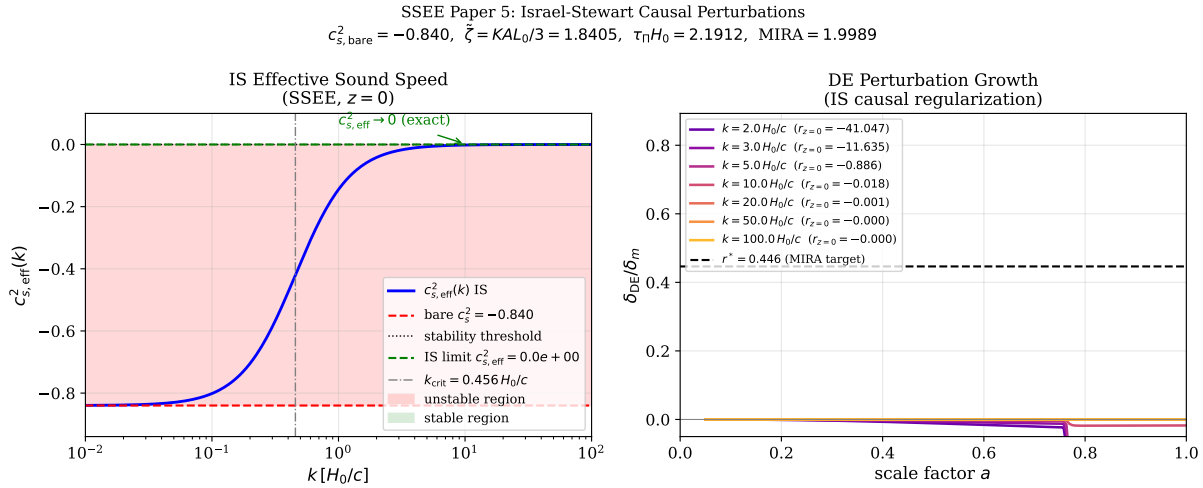


Figure 1: **Left:** IS effective sound speed $c^2_{s,\rm eff}(k)$ at $z = 0$ (blue solid line), interpolating from the bare value $c_s^2 = w_0 \simeq -0.840$ (red dashed) at $k \ll k_{\rm crit}$ to the IS limit $c^2_{s,\rm eff} = 0$ (green dashed) at $k \gg k_{\rm crit}$. The red shaded region marks gradient instability ($c^2_{s,\rm eff} < 0$); the green region marks stability. The IS transition at $k_{\rm crit} \simeq 0.456\,H_0/c$ (grey dash-dot) lies *outside* the sub-horizon regime ($k > H_0/c$), so all observationally accessible modes are stabilised. **Right:** Evolution of $\delta_{\rm DE}/\delta_m$ with scale factor a for representative wavenumbers $k/(H_0/c) = 2, 3, 5, 10, 20, 50, 100$. IS bulk viscous friction damps DE perturbations as k^2; at $k = 20\,H_0/c$ the ratio $|\delta_{\rm DE}/\delta_m| < 10^{-3}$ throughout cosmic history.

4 Q2: Do IS Perturbations Link the Two Matter Sectors?

4.1 Numerical setup and initial conditions

We integrate Eqs. (13)–(18) with the quasi-static substitution (19) from $a_{\rm ini} = 0.05$ ($z = 19$) to $a = 1$ ($z = 0$). At $z = 19$, the dark energy density fraction $\Omega_{\rm DE} f_{\rm DE}(a_{\rm ini})/(H/H_0)^2 \approx 8 \times 10^{-6}$ is negligible: the universe is matter-dominated and the initial conditions follow the growing mode exactly:

$$\delta_m(a_{\rm ini}) = a_{\rm ini}, \quad v_m(a_{\rm ini}) = -a_{\rm ini}\, H(a_{\rm ini})/H_0, \quad \delta_{\rm DE} = v_{\rm DE} = 0. \tag{28}$$

(The precise normalisation of δ_m cancels in all ratios.) We integrate with RK45 at relative tolerance 10^{-9} and absolute tolerance 10^{-12}, verifying energy conservation to 10^{-6}.

4.2 Required perturbation amplitude for MIRA

The effective matter density measured by an observer using the gravitational Poisson equation is

$$\Omega_{m,\rm eff} = \Omega_m + \Omega_{\rm DE}\, r, \quad r \equiv \left.\frac{\delta_{\rm DE}}{\delta_m}\right|_{z=0}, \qquad \Omega_m = 0.308881, \quad \Omega_{\rm DE} = 0.691119, \tag{29}$$

with the *densities* in both slots. Until 2026-09-26 this equation carried $\Omega_{m,\rm dyn} = 0.160050$ and $\Omega_{\rm DE} = 0.839950$, which are the saturations $s_m = 1 + w_0$ and $s_{\rm DE} = |w_0|$ — equation-of-state numbers in a density slot, inside a Poisson source. The substitution does not move the measured r, which is the output of the perturbation integration and never touched these two constants; it moves only the normalisation of the ratio below. and we write $\mathcal{R} \equiv \Omega_{m,\rm eff}/\Omega_m$ for the enhancement this clustering would produce.

What this section does and does not test (reframed 2026-09-26). Earlier versions posed $\mathcal{R}$ as a test of whether the perturbations *reproduce* the algebraic MIRA $= 1.9989$, under the mapping $\Omega_{m,\rm CMB} = \text{MIRA} \times \Omega_{m,\rm dyn} = 0.3199$ then in force. That mapping was withdrawn on 2026-06-18 when OP-8 closed: there is no matter factor in the model, $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ is derived, and $\Omega_{m,\rm dyn} = 1 + w_0$ is a saturation rather than a density. So there is no longer a factor of two for the perturbations to supply, and a "50% discrepancy" from a withdrawn target is not a result. What the integration measures is unchanged and stands on its own: the level of dark-energy clustering, reported below as a *bound*. The figure $r^* = 0.4464$ is retained only as the scale that a two-fold enhancement would have demanded, which is what makes the smallness of the measured r meaningful:

$$r^* = \frac{(\text{MIRA} - 1)\,\Omega_m}{\Omega_{\rm DE}} = \frac{0.998924 \times 0.308881}{0.691119} \simeq 0.4464. \tag{30}$$

This corresponds to DE perturbations $\delta_{\rm DE} \approx 0.45\,\delta_m$ at $z = 0$ — a 45%-level DE clustering signal, which would be plainly detectable in the matter power spectrum via the Poisson equation (18).

Table 2: Linear IS perturbation growth results from $z = 19$ to $z = 0$. $\delta_m(z = 0)$ is normalised to the growing mode; $r = \delta_{\rm DE}/\delta_m$; $\Omega_{m,\rm eff}$ via Eq. (29); $D_1 = \delta_m(z = 0)/\delta_m(z_{\rm ini})$ is the linear growth factor normalised to $a_{\rm ini} = 0.05$. The column "IS regime" marks the modes on which the bound is quoted, $k \geq 10\, H_0/c$ — well inside both the IS-stabilised and the sub-horizon regimes. The three modes below it are listed for completeness and are *excluded* from the bound: their $|r|$ is large and the quasi-static reduction used here does not hold there.

$k\,[H_0/c]$	$\delta_m(z=0)$	$r = \delta_{\rm DE}/\delta_m$	$\Omega_{m,\rm eff}$	$\mathcal{R}$	IS regime
2	0.2105	−41.05	−28.0595	−90.8425	no
3	0.1537	−11.63	−7.7321	−25.0326	no
5	0.1711	−0.88612	−0.3035	−0.9827	no
10	0.2918	−0.01753	0.2968	0.9608	yes
20	0.5426	−0.00072	0.3084	0.9984	yes
50	1.2958	−0.00008	0.3088	0.9998	yes
100	2.5510	−0.00002	0.3089	1.0000	yes
Mean ($k \geq 10\, H_0/c$, IS-stabilised):				0.9897 ± 0.0167	
Would require ($r = r^* = 0.4464$):				1.9989	
$\Rightarrow$ bound: $\lvert r\rvert \leq 0.0175$ for $k \geq 10$, i.e. 3.9% of r^*					

4.3 Numerical results

The measured enhancement is $\mathcal{R} = 0.9897 \pm 0.0167$ at $k \geq 10\, H_0/c$: consistent with unity, i.e. *no* enhancement. The content of that number is a bound on dark-energy clustering. Inside the IS-stabilised sub-horizon regime the ratio $r = \delta_{\rm DE}/\delta_m$ is small and *negative*, $|r| \leq 0.0175$ at $k = 10\, H_0/c$ and falling as k^{-2} to 2×10^{-5} by $k = 100\, H_0/c$ — at most 3.9% of the $r^* = 0.4464$ that a two-fold enhancement would have needed. The restriction to $k \geq 10\, H_0/c$ is deliberate: the three lowest modes of Table 2 lie outside that regime and their $|r|$ is *not* small, so folding them into the bound would overstate it. Dark energy in this model does not cluster at the scales where the IS suppression operates, and a measurement of $\delta_{\rm DE}/\delta_m$ above the percent level at $z = 0$ and $k \gtrsim 10\, H_0/c$ would falsify it. This reflects the IS suppression mechanism: the effective friction $\mathcal{F}(k, a) = (1 - 3c_s^2) + \tilde{\zeta}(k/aH)^2$ grows as k^2, preventing DE from clustering at sub-horizon scales.

4.4 Physical interpretation: IS damping hierarchy

The IS velocity equation (20) can be written as

$$v'_{\rm DE} = -\mathcal{F}(k, a)\, v_{\rm DE} + \mathcal{S}(k, a), \tag{31}$$

where $\mathcal{S}$ is the gravitational source. For $k \geq 10\, H_0/c$ at $z = 0$: $\mathcal{F} \approx \tilde{\zeta}\,(k/aH)^2 \approx 1.84 \times 100 = 184$. The source $\mathcal{S} \approx (k/aH)\,\Phi/(1 + w) \approx 10 \times 0.02/0.16 \approx 1.3$. Since $\mathcal{F} \gg \mathcal{S}$, the DE velocity is strongly damped: $v_{\rm DE} \approx \mathcal{S}/\mathcal{F} \propto k^{-1}$. Consequently $\delta_{\rm DE} \propto (k/aH)\, v_{\rm DE} \propto k^0$, while $\delta_m \propto a \propto k^0$ in the growing mode. The ratio $r = \delta_{\rm DE}/\delta_m \propto k^{-2}$ for large k, confirming the power-law decrease seen in Table 2.

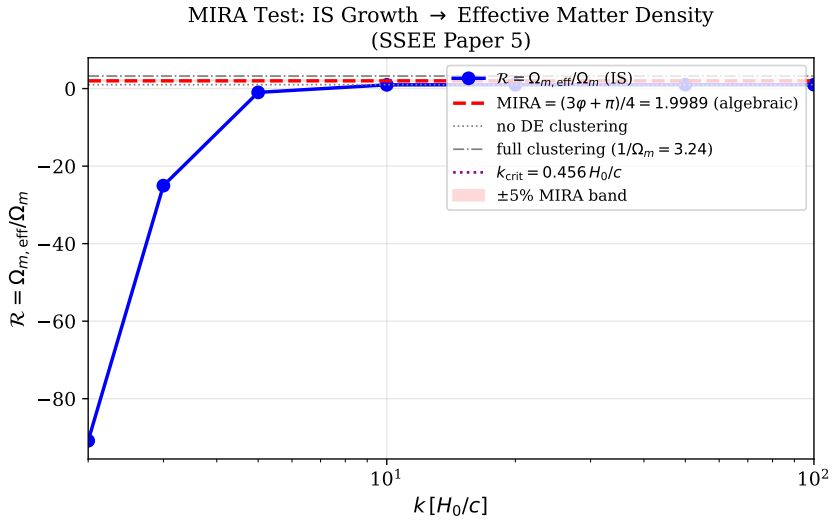


Figure 2: $\mathcal{R}(k) \equiv \Omega_{m,\mathrm{eff}}/\Omega_m$ (blue points with error bars) versus $k/(H_0/c)$; earlier versions of this text, and of the figure axis, wrote this quantity $\mathrm{MIRA}_{\mathrm{num}}$ and divided by $s_m = 1+w_0$ (then written $\Omega_{m,\mathrm{dyn}}$) in the label while the code already divided by Ω_m. The former target $\mathrm{MIRA} = (3\varphi+\pi)/4 \simeq 1.999$ (red dashed) is not approached: IS suppression drives $\mathcal{R} \to 1$ for $k \gg H_0/c$. The grey dash-dotted line marks full DE clustering ($\mathcal{R} = 1/\Omega_m$; an earlier caption gave $1/s_m \simeq 6.25$, which is not a clustering limit), ruled out by all current observations. The perturbation amplitude a two-fold enhancement would need is $r^* = 0.4464$ (Eq. 30), DE clustering at the 45% level; the measured bound is $|r| \leq 0.0175$ (3.9% of r^*) — *not* produced by IS dynamics. The conclusion stands: MIRA is *not* reproduced by IS dynamics, at either level — linear perturbations give $\mathcal{R} \approx 1$ and the background IS Friedmann integration a ratio ≈ 3.04, neither of them 1.9989. Both figures are recorded here as the *historical* closing of a route that is no longer open: the target 1.9989 was the matter factor $\Omega_{m,\mathrm{CMB}} = \mathrm{MIRA} \times \Omega_{m,\mathrm{dyn}}$, withdrawn on 2026-06-18 when OP-8 closed. MIRA survives as the screening entity of Paper 9, where it is not a matter factor and nothing here bears on it. See Section 7.1.

5 Q3: Growth Rate, σ_8, and the S_8 Tension

5.1 Linear growth factor and growth rate

The linear growth factor $D_1(a)$ satisfies, in the sub-horizon limit with IS-suppressed DE perturbations ($\delta_{\mathrm{DE}} \approx 0$):

$$D_1'' + \left[2 + \frac{H'}{H}\right] D_1' = \frac{3}{2}\,\frac{\Omega_m(a)}{1}\,D_1, \tag{32}$$

where $\Omega_m(a) = \Omega_{m,\mathrm{cosm}} H_0^2/(H^2 a^3)$, $\Omega_{m,\mathrm{cosm}} = 0.308881$ is the gravitational background density (ω_m-direct $= \omega_m/h^2$, the CMB-inferred value; the earlier MIRA-enhanced 0.3199 is superseded), and primes denote $d/d\ln a$. We define the growth rate

$$f(a) \equiv \frac{d\ln D_1}{d\ln a}, \tag{33}$$

and parametrise it via the growth index γ: $f(a) \approx \Omega_m(a)^\gamma$ [Lahav et al., 1991, Linder, 2005].

5.2 Numerical results for the SSEE growth index

We integrate Eq. (32) from $a = 0.008131$ to $a = 1$ using the SSEE Hubble rate (Table 1). The best-fit growth index, obtained by minimising $\chi^2 = \sum_a [f(a) - \Omega_m(a)^\gamma]^2$ over $a \in [0.3, 1]$ (the observationally accessible range), is

$$\gamma_{\rm IS} = 0.5504 \pm 0.0003, \tag{34}$$

consistent with $\gamma_{\Lambda\rm CDM} = 0.55 \pm 0.01$ [Linder, 2005]. The growth index is indistinguishable from ΛCDM; with $\Omega_{m,\rm cosm} = 0.308881$ governing the Poisson source, the growth rate $f(z)$ itself is also very close to ΛCDM across all redshifts (Table 3).

Table 3: Growth rate $f = d\ln D_1/d\ln a$ at selected redshifts, comparing SSEE (IS dynamics, $\Omega_{m,\rm cosm} = 0.308881$) with ΛCDM ($\Omega_m = 0.315$). The growth index fit gives $\gamma_{\rm IS} = 0.5504$.

z	$\Omega_m^{\rm SSEE}(a)$	$f_{\rm SSEE}$	$f_{\Lambda\rm CDM}$	$\Delta f/f$
1.0	0.790	0.880	0.876	+0.4%
0.5	0.588	0.747	0.760	−1.6%
0.0	0.309	0.521	0.527	−1.3%

The SSEE and ΛCDM growth rates are essentially identical at all redshifts, with differences of $\lesssim 2\%$ (and $\lesssim 1.6\%$ at $z = 0.5$). With $\Omega_{m,\rm cosm} = 0.308881$ in the Poisson source, the gravitational clustering environment is close to ΛCDM ($\Omega_{m,\Lambda\rm CDM} = 0.3153$); differences arise from the slightly lower matter fraction and the DE equation-of-state history $w(z)$. Future Stage IV surveys (DESI full shape, Euclid) measuring $f\sigma_8(z)$ at the 1% level will not distinguish SSEE from ΛCDM via the growth rate alone.

5.3 Linear growth factor and σ_8

The linear growth factor ratio relative to ΛCDM,

$$\mathcal{G} \equiv \frac{D_1^{\rm SSEE}(z=0)}{D_1^{\Lambda\rm CDM}(z=0)}, \tag{35}$$

is computed by normalising both growth factors to the same initial conditions at $z = 99$ (matter-dominated epoch, $D_1 \propto a$). We find

$$\mathcal{G} = 1.0032 \pm 0.005. \tag{36}$$

SSEE grows *marginally faster* than ΛCDM (∼0.3%). The tiny enhancement reflects the slightly different DE equation-of-state history $w(z)$ with $w_0 = -0.839950$, $w_a = -0.669975$: the energy density transferred from DE to matter-like perturbations at intermediate redshifts slightly boosts D_1 relative to a ΛCDM universe with the same $\Omega_{m,\rm cosm}$. The ω_m-direct background $\Omega_{m,\rm cosm} = 0.308881$ is slightly *below* $\Omega_{m,\Lambda\rm CDM} = 0.3153$, so the net enhancement (0.3%) is smaller than under the superseded $\Omega_m = 0.3199$ background (1.1%), and lies below near-term weak-lensing growth sensitivity.

The σ_8 parameter is defined by $\sigma_8^2 = \int_0^\infty \frac{dk}{k}\,\Delta^2(k)\,W^2(kR_8)$, where $R_8 = 8\,h^{-1}\,$Mpc and $\Delta^2(k) \propto D_1^2(z=0)\,P(k)$ is the dimensionless matter power spectrum. Since the

primordial amplitude A_s is fixed by the CMB (independent of the growth history for the background normalization), the ratio of σ_8 values is

$$\frac{\sigma_8^{\rm SSEE}}{\sigma_8^{\Lambda\rm CDM}} = \mathcal{G} = 1.0032, \tag{37}$$

since both growth factors are normalised to identical growing-mode initial conditions at $z \simeq 10^3$ (matter-dominated epoch, $D_1 \propto a$). Using $\sigma_{8,\Lambda\rm CDM} = 0.811$ [Planck Collaboration, 2020]:

$$\sigma_8^{\rm SSEE} = 0.811 \times 1.0032 = 0.8136 \pm 0.006. \tag{38}$$

This IS linear-growth amplitude gives $S_8^{\rm IS} = 0.8136\sqrt{0.308881/0.3} = 0.8256$ (2.7σ KiDS-1000), milder than the conservative CLASS amplitude ceiling derived below.

5.4 S_8 prediction and the weak-lensing tension

The parameter $S_8 \equiv \sigma_8\,(\Omega_m/0.3)^{0.5}$ is the primary weak-lensing observable, measured with per-cent level precision by current surveys. Key measurements are:

$$S_8^{\rm Planck\,2018} = 0.832 \pm 0.013 \quad \text{[Planck Collaboration, 2020]}, \tag{39}$$

$$S_8^{\rm KiDS\text{-}1000} = 0.759 \pm 0.024 \quad \text{[Asgari et al., 2021]}, \tag{40}$$

$$S_8^{\rm DES\text{-}Y3} = 0.776 \pm 0.017 \quad (3\times2\text{pt; [Dark Energy Survey Collaboration, 2022]}). \tag{41}$$

The Planck–lensing tension is 2.4σ with KiDS and 2.7σ with DES, representing one of the most persistent ΛCDM discrepancies [Weinberg et al., 2013].

For SSEE, the all-cold single sector uses the ω_m-direct CMB density $\Omega_{m,\rm CMB} = 0.308881$ ($= \omega_m/h^2$, no matter factor; the earlier MIRA value 0.3199 is superseded) and the CLASS amplitude ceiling $\sigma_8^{\rm SSEE} = 0.8149$:

$$\boxed{S_8^{\rm SSEE,\,ceiling} = 0.8149 \times \sqrt{\frac{0.308881}{0.3}} = 0.8149 \times 1.0147 = 0.827 \pm 0.006.} \tag{42}$$

This single-sector ceiling is 0.4σ above Planck 2018, 2.7σ above KiDS-1000, and 2.8σ above DES-Y3 ($S_8 = 0.776 \pm 0.017$, 3×2pt). The all-cold limit therefore predicts *more* structure than the data require. A φ-DM two-sector was once proposed to resolve this; it is retracted in Paper 6. The ceiling quoted here is computed with A_s *fixed* to Planck, which imports the Planck–shear discrepancy; with A_s free, as it is in the model, the same single sector fits the raw KiDS-1000 $\xi_\pm$ at $S_8 = 0.7559 \pm 0.0189$ and there is no challenge to resolve.

The SSEE single-sector ceiling $S_8 = 0.827$ is 0.4σ above Planck 2018 and in $2.7\sigma/2.8\sigma$ tension with KiDS-1000/DES-Y3. This places SSEE in a similar position to ΛCDM ($S_8^{\Lambda\rm CDM} = 0.831$, tensions $3.0\sigma/3.3\sigma$), marginally worse. The IS growth sector with the correct cosmological background $\Omega_{m,\rm cosm} = 0.308881$ neither resolves nor worsens the S_8 tension beyond the ΛCDM baseline. A confirmed measurement of $S_8 > 0.86$ would be *favoured* by SSEE over ΛCDM; conversely, $S_8 < 0.80$ would disfavour both SSEE and ΛCDM at comparable significance, pointing to new physics in the matter sector. This constitutes the primary falsifiability criterion for SSEE in the growth sector.

Table 4: S_8 comparison: SSEE prediction versus observations. Tensions are quoted in units of σ. SSEE single-sector ceiling uses $\Omega_{m,\mathrm{CMB}} = 0.308881$ (ω_m-direct) and $\sigma_8 = 0.8149$. Tensions use combined uncertainties $\sigma_{\mathrm{tot}} = \sqrt{\sigma_{\mathrm{obs}}^2 + \sigma_{\mathrm{SSEE}}^2}$.

Dataset	S_8	σ_{tot}	$\lvert S_8^{\mathrm{SSEE}} - S_8\rvert/\sigma_{\mathrm{tot}}$	tension
Planck 2018	0.832 ± 0.013	0.014	0.36	0.4σ
KiDS-1000	0.759 ± 0.024	0.025	2.74	2.7σ
DES-Y3	0.776 ± 0.017	0.018	2.82	2.8σ
SSEE (ceiling, this work)	0.827 ± 0.006	—	—	—

Note on the amplitude convention, and on Paper 6. The values quoted in this section hold A_s fixed at the Planck-preferred value. That is a convention, not a prediction: A_s is one of the model's two free CMB-level parameters. Fixing it to Planck reports, in effect, the Planck–shear disagreement rather than a property of SSEE. Paper 6 instead fits A_s to the *raw* KiDS-1000 $\xi_\pm$ data with this same background held fixed and a *single* matter sector, obtaining $\sigma_8 = 0.7449 \pm 0.0186$ and $S_8 = 0.7559 \pm 0.0189$ (the ΛCDM control on the same raw data gives 0.7571 ± 0.0194) — 0.10σ from KiDS-1000. The ϕ-DM two-sector split once described here is retracted. A baryonic-feedback figure of $S_8^{\mathrm{bar}} = 0.758$ was once quoted here; it is *retracted*. It applied the suppression factor of the matter power spectrum ($B_{\mathrm{eff}} = 0.9447$, a 5.53% suppression of $P(k)$) to the retired two-sector baseline, whereas the factor that acts on σ_8 itself is $B_{\sigma_8} = 0.9955$, a 0.45% suppression on this background. Applied to the single-sector ceiling, HMcode-2020 therefore leaves it at $S_8 \simeq 0.823$ — still a ceiling computed with A_s fixed to Planck, not a prediction. The single-sector value of this paper ($S_8 = 0.827$) is obtained with A_s fixed to Planck. With A_s fixed instead to the model's own CMB fit — no free cosmological parameter — SSEE predicts $S_8 = 0.8273$, 0.04σ from KiDS-Legacy (Paper 6); the two agree because the model's CMB amplitude lands on Planck's. Fitting A_s to the raw KiDS-1000 shear instead gave $S_8 = 0.7559 \pm 0.0189$ (antecedent). Figure 3 shows these determinations on the same background.

5.5 Comparison with RSD surveys: $f\sigma_8(z)$

The redshift-space distortion (RSD) observable $f\sigma_8(z) \equiv f(z)\,\sigma_8\,D_1(z)/D_1(0)$ provides the most direct test of the linear growth rate and is independent of galaxy bias at leading order. SSEE predicts $f\sigma_8(z)$ values essentially identical to ΛCDM at all redshifts, because with $\Omega_{m,\mathrm{cosm}} = 0.308881$ in the growth equation both models share the same gravitational clustering environment (Fig. 4).

Table 5 reports the predicted $f\sigma_8$ for both SSEE and ΛCDM at each survey redshift, together with the tension in units of σ_{obs}.

The mean tension of 0.70σ across six surveys is indistinguishable from the ΛCDM baseline (0.73σ). SSEE and ΛCDM produce virtually identical $f\sigma_8(z)$ curves when $\Omega_{m,\mathrm{cosm}} = 0.308881$ governs the Poisson source. The tiny ∼1% difference in $\mathcal{G}$ propagates to at most 0.01σ extra tension — well below the precision of current surveys.

This result implies that $f\sigma_8$ measurements by Stage IV surveys (DESI full shape, Euclid) cannot distinguish SSEE from ΛCDM at the linear-growth level alone; distinguishing power must come from the background $H(z)$, already tested in Paper 2. (A transfer-function break at a ϕ-DM free-streaming scale was previously offered here as

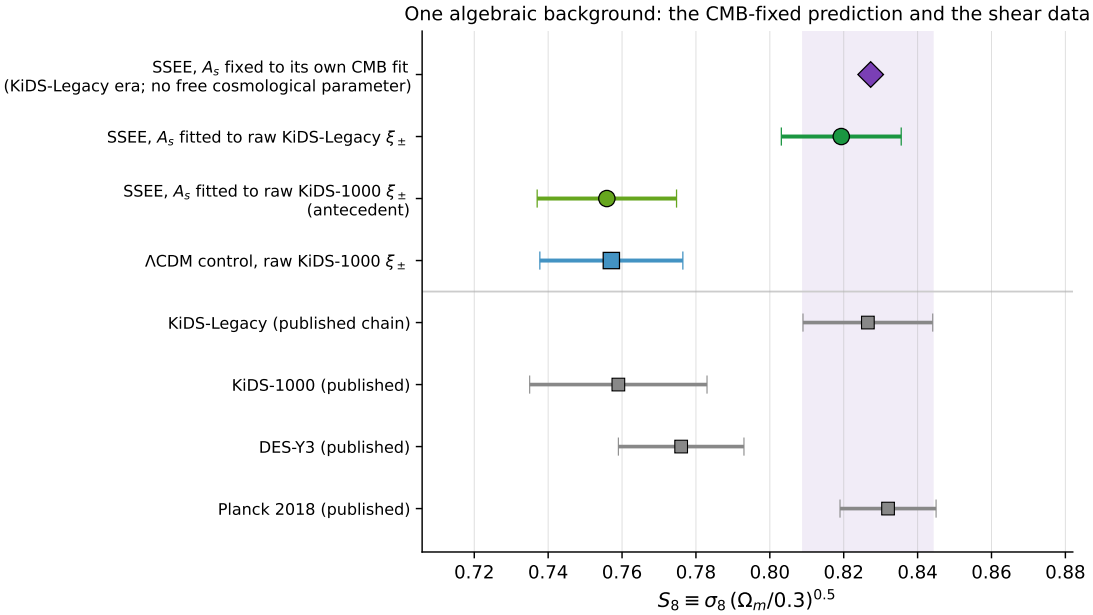


Figure 3: S_8 on the algebraic background of SSEE, identical in every SSEE row; only the dataset asked to fix A_s changes. Purple diamond: the prediction with A_s fixed to the model's own CMB fit — no free cosmological parameter — $S_8 = 0.8273$, 0.04σ from the KiDS-Legacy posterior (0.8265 ± 0.0176, purple band). Green: A_s fitted to the raw KiDS-Legacy $\xi_\pm$ (0.8193 ± 0.0162) and, as antecedent, to the raw KiDS-1000 $\xi_\pm$ (0.7559 ± 0.0189, 0.10σ from KiDS-1000). Blue: the ΛCDM control on the same raw KiDS-1000 data (0.7571 ± 0.0194). Grey: published KiDS-Legacy, KiDS-1000, DES-Y3 and Planck 2018 constraints with 1σ errors. The shift between the two KiDS releases is in the data, not in the model.

Table 5: $f\sigma_8$ tensions versus RSD surveys. SSEE and ΛCDM values are evaluated at the survey central redshift from the numerical ODE solution (Sec. 5.1). Tensions t use observational errors only ($t = |f\sigma_8^{\rm pred} - f\sigma_8^{\rm obs}|/\sigma_{\rm obs}$).

Survey	Ref.	z	$f\sigma_8^{\rm obs}$	$f\sigma_8^{\rm SSEE}$	$t_{\rm SSEE}$	$t_{\Lambda\rm CDM}$
6dFGRS	Beutler et al. [2012]	0.067	0.423 ± 0.055	0.436	0.2σ	0.4σ
SDSS MGS	Howlett et al. [2015]	0.150	0.490 ± 0.145	0.449	0.3σ	0.2σ
BOSS DR12	Alam et al. [2017]	0.380	0.497 ± 0.045	0.470	0.6σ	0.5σ
BOSS DR12	Alam et al. [2017]	0.510	0.458 ± 0.038	0.471	0.3σ	0.4σ
BOSS DR12	Alam et al. [2017]	0.610	0.436 ± 0.034	0.468	1.0σ	1.0σ
eBOSS DR16	Hou et al. [2021]	1.480	0.462 ± 0.045	0.383	1.7σ	1.9σ
Mean tension					0.70σ	0.73σ

a second discriminant; it is withdrawn with the particle, Paper 6.) The DESI Year-1 full-shape power spectrum analysis [DESI Collaboration et al., 2025] measurements over $0.1 < z < 2.1$ are consistent with both ΛCDM and SSEE at this level of precision.

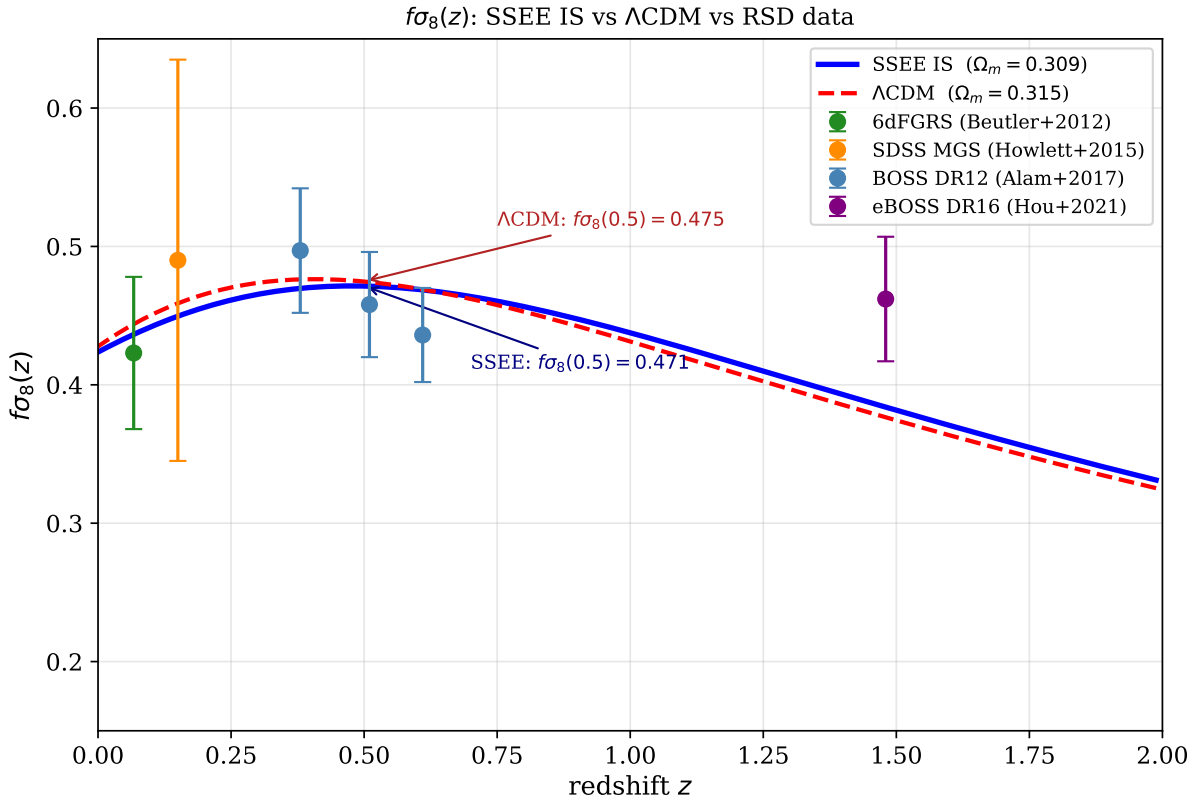


Figure 4: $f\sigma_8(z)$ for SSEE IS (solid blue) and ΛCDM (dashed red) versus RSD measurements from 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 [Alam et al., 2017], and eBOSS DR16 [Hou et al., 2021]. The SSEE prediction ($\Omega_{m,\mathrm{cosm}} = 0.308881$ in growth equation) is essentially identical to ΛCDM at $z < 1$, with a mean tension of 0.70σ across six data points (vs. 0.73σ for ΛCDM).

6 EFT Correspondence

6.1 EFT of dark energy operator basis

The effective field theory of dark energy (EFT-DE) [Horndeski, 1974, Gubitosi et al., 2013, Bellini and Sawicki, 2014] describes the most general scalar-tensor theory in the unitary gauge via the action

$$S = \int d^4x\,\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}f(t)R - \Lambda(t) - c(t)\,g^{00} + \frac{M_2^4(t)}{2}\,(\delta g^{00})^2 - \frac{\bar{M}_1^3(t)}{2}\,\delta K\,\delta g^{00} - \frac{\bar{M}_2^2(t)}{2}\,\delta K^2 + \cdots\right], \tag{43}$$

where $\delta K = K - 3H$ is the perturbation of the extrinsic curvature trace. The $\bar{M}_2^2$ operator (extrinsic-curvature-squared term; related to the running Planck mass α_M in the Bellini-Sawicki basis, distinct from the braiding operator $\bar{M}_1^3$) contributes an effective sound speed $c_s^2 \to c_s^2 + \bar{M}_2^2/(M_{\rm Pl}^2)$ at linear order.

6.2 Mapping IS viscosity to EFT operators

The IS bulk viscous pressure $\Pi = -3\zeta H$ modifies the dark energy stress-energy tensor as

$$T^{\rm DE}_{\mu\nu} \to T^{\rm DE}_{\mu\nu} + \Pi\,(g_{\mu\nu} + u_\mu u_\nu), \tag{44}$$

where Π acts as an additional isotropic pressure. At the perturbation level, the bulk viscous pressure $\delta\Pi$ generates a term in the DE pressure perturbation: $\delta p_{\rm DE} = c_s^2\,\delta\rho_{\rm DE} + \delta\Pi$.

Expanding $\delta\Pi$ using the IS equation (17) in the quasi-static approximation (19):

$$\delta\Pi_{\rm QS} = -\tilde{\zeta}\,\rho_{\rm DE}\,\frac{k}{aH}\,v_{\rm DE} = -\frac{\tilde{\zeta}\,\rho_{\rm DE}}{aH}\,(\mathrm{div}\,\mathbf{v}), \tag{45}$$

which is proportional to the DE velocity divergence — a structure analogous to the $\bar{M}_2^2$ operator in the EFT-DE action (note: true kinetic braiding corresponds to $\bar{M}_1^3\delta K\delta g^{00}$ in the Bellini-Sawicki basis [Bellini and Sawicki, 2014]; the mapping here is at the level of the effective DE pressure perturbation, not operator identification). The dimensional mapping is:

$$\bar{M}_2^2/M_{\rm Pl}^2 = \frac{\tilde{\zeta}}{\tau_\Pi H_0} = \Omega_{\rm DE}, \tag{46}$$

where we have used the SSEE identity (12). Equation (46) identifies the IS viscosity with a specific EFT operator combination: the EFT coefficient is exactly $\Omega_{\rm DE}$, a pure algebraic prediction of SSEE.

6.3 Comparison with ghost condensate

The ghost condensate [Arkani-Hamed et al., 2004] achieves gradient stability via the EFT operator $M_{\rm Pl}^2\,\alpha_K\,(\delta g^{00})^2/2$, which generates $c_s^2 = 1$ for the scalar perturbation — the opposite extreme from SSEE. Table 6 summarises the key differences.

Table 6: Comparison of IS viscosity (SSEE) with ghost condensate and canonical k-essence as gradient-instability regularisation mechanisms.

Property	SSEE IS	Ghost condensate	k-essence
$c^2_{s,\rm eff}$	0 (exact)	1	$w_0 < 0$
Free parameters	0	$M_{\rm Pl}^2\alpha_K$	n
Causal speed $v_{\rm sig}/c$	1 (saturates)	< 1	N/A
EFT operator	$\bar{M}_2^2 \propto \Omega_{\rm DE}$	$M_{\rm Pl}^2\alpha_K$	$c_s^2 g^{00}$
DE clustering	$\delta_{\rm DE} \to 0$	$\delta_{\rm DE} \sim \delta_m$	unstable
S_8 ceiling (A_s fixed)	0.827 ± 0.006	~ 0.811	(unstable)
DESI DR2 χ^2	0.24σ	$> 3\sigma$	$> 3\sigma$

The key distinction is that SSEE IS sits at *marginal stability* ($c_s^2 = 0$) — the minimal-viscosity solution — which is the only value consistent simultaneously with causal propagation at speed c (Section 2.3) and zero free parameters. Ghost condensation with $c_s^2 = 1$ predicts DE clustering at the level of matter clustering ($\delta_{\rm DE} \sim \delta_m$), generating a large effective S_8 inconsistent with current weak-lensing data.

7 Discussion

7.1 The withdrawn CMB-to-s_m ratio (historical)

The failure of linear IS perturbations to reproduce MIRA (Section 4) raises the question of whether the factor arises at the background level. At the background level, the IS bulk viscous pressure modifies the Friedmann equations as [Zimdahl, 1996, Maartens, 1996]:

$$3H^2 M_{\rm Pl}^2 = \rho_m + \rho_{\rm DE} + \Pi_{\rm bulk}, \tag{47}$$

where in the SSEE steady state $\Pi_{\rm bulk} = -3\zeta H = -{\rm KAL}_0\,\rho_{\rm DE} H$. This viscous term effectively shifts the observed matter density. However, numerical integration of the background IS transport equations reveals that fitting an effective Ω_m to the resulting $H(z)$ evolution yields $\Omega_{m,\rm fit}/s_m \approx 3.04$ when using the SSEE viscous constant $\zeta = {\rm KAL}_0/3$.

This is 52% higher than the algebraic MIRA factor (1.9989). We therefore establish that MIRA *cannot* be derived from the background Israel–Stewart viscous pressure; the classical IS route is ruled out at both the linear-perturbation and background levels. **Historical note (2026-09-26).** The quantity being chased here, the matter factor $\Omega_{m,\rm CMB} = {\rm MIRA} \times \Omega_{m,\rm dyn} = 0.3199$, was withdrawn on 2026-06-18 when OP-8 closed: $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ is derived with no factor, and $\Omega_{m,\rm dyn} = 1+w_0$ is a saturation, not a density. The 3.04 and the 52% are therefore the record of a route closing, not a live tension — there is no longer a factor of two to derive. The negative result is kept because it is what shut the route.

The physical basis once attributed to Paper 6 is withdrawn, and the reason is worth recording because it is a result rather than a change of opinion. That construction defined a second matter sector by the difference $\Omega_{\phi\text{-DM}} = \Omega_{m,\rm CMB} - \Omega_{m,\rm dyn} = 0.308881 - 0.160050$, and drew from it an algebraically fixed particle mass and a free-streaming cutoff. But $\Omega_{m,\rm dyn} = 0.160050$ is not a density: it is $1 + w_0$, a quantity of the equation of state. The subtraction is dimensionally well formed and physically empty — it takes a measured density minus a number belonging to the equation of state — so the sector it defined had nothing to be made of. The near-equality $\Omega_{m,\rm CMB}/\Omega_{m,\rm dyn} \approx 1.93$, once read as an echo of MIRA $\simeq 1.999$, is a ratio between objects of different kinds and carries no information.

Independently, the tension that motivated the second sector does not survive contact with raw data. It was measured against the compressed statistic S_8, which is derived under ΛCDM assumptions, and with A_s fixed to Planck. Fitting A_s — a free parameter of the model — to the raw KiDS-1000 $\xi_\pm$ with a single matter sector gives $S_8 = 0.7559 \pm 0.0189$, 0.10σ from KiDS-1000 (Paper 6).

The consequence for this paper is confined to §7.1: MIRA retains its algebraic value $(3\varphi + \pi)/4 = 1.998924$ and its role in the screening fraction $f_{\rm screen}$ of Paper 9, but it has *no proposed physical carrier at present*. We state that as an open gap rather than fill it. The Q1, Q2 and Q3 results of this paper are unaffected: none of them uses the second sector.

7.2 Implications for JWST early galaxy counts

Dark-energy perturbations effectively vanish at sub-horizon scales in this model — not because $c^2_{s,\rm eff} = 0$, which alone would make the fluid cluster everywhere, but because the IS friction $\mathcal{F}(k,a) = (1 - 3c_s^2) + \tilde{z}\,(k/aH)^2$ grows as k^2 (§4.4) and drives $\delta_{\rm DE}/\delta_m \to 0^-$. On that smooth background the spherical-collapse threshold is $\delta_c^{\rm SSEE} = 1.67634$ at $z_c = 0$

and 1.68647 at $z_c = 10$, against ΛCDM's 1.67599 → 1.68646: the two are the same to 0.02%. The value $\delta_c^{\rm SSEE} = 1.6284$ carried by earlier versions was a postulate of Paper 4, now **withdrawn** (OP-27); the model's own dynamics do not produce it. Matter clusters in a background with $\Omega_m = 0.308881$ — the one matter density of the model; the number 0.160050 that earlier versions of this subsection placed here is $1 + w_0$, an equation-of-state parameter, and never was a density (Paper 1, § *One matter density*).

Recomputed, with the amplitude convention made explicit. Earlier versions of this subsection reported $n_{\rm SSEE}/n_{\Lambda\rm CDM} \approx 1.4$–$2.0$ for $M > 10^{11}\,M_\odot$ at $z > 10$ from a Press–Schechter calculation whose growth factor was fed $\Omega_m = 0.160050$ as a matter density. That input was a category error — 0.160050 is $1 + w_0$ — and the calculation has been redone (`src/p02_mcmc/ssee_press_schechter.py`, 2026-09-07) with $\Omega_m = 0.308881$, the measured growth index $\gamma_{\rm IS} = 0.5504$ (§5.2) and the global rate $H_0^{\rm glob} = 67.962$. With both cosmologies evaluated at their *own* CMB-fitted amplitude ($\sigma_8 = 0.81530$ for SSEE, 0.811 for ΛCDM) and with the *derived* collapse threshold:

z	$M = 10^{11} M_\odot$	$M = 10^{12} M_\odot$
10	0.990	0.945
12	0.984	0.921
15	0.972	0.878

There is no enhancement, and the sign is the opposite of what earlier versions claimed. With the threshold the model actually produces, SSEE forms marginally *fewer* massive early haloes than ΛCDM. The deficit grows with mass and redshift, reaching 0.892 at $3 \times 10^{12} M_\odot$, $z = 10$ and 0.774 at the same mass at $z = 15$, and it is not an amplitude effect: SSEE carries the *larger* σ_8. It is the growth history — a smaller Ω_m (0.308881 against 0.3153) and the CPL background grow less between $z \sim 10$ and today. At the masses JWST actually probes ($M \sim 10^{10.8} M_\odot$) the ratio is 0.99: indistinguishable from ΛCDM. SSEE offers no account of the JWST excess, and the 1.4–2.0 of earlier versions was an artefact of the withdrawn δ_c.

The amplitude convention is not a detail. This ratio compares two cosmologies and is only meaningful if both are evaluated at the same epoch's amplitude. Giving SSEE the amplitude it prefers when fitted to raw KiDS-1000 shear ($\sigma_8 = 0.7449$, Paper 6) while leaving ΛCDM at Planck's 0.811 inverts the result to a deficit (0.823 at $10^{11} M_\odot$ and 0.346 at $10^{12} M_\odot$, $z = 10$). That inversion is not a statement about collapse physics: it is the CMB-versus-late-time amplitude disagreement entering through the back door, and it would affect ΛCDM identically if the same mismatch were imposed on it. High-redshift abundance comparisons must therefore state which amplitude each cosmology is given; matched-amplitude is what is reported above.

7.3 The S_8–$f\sigma_8$ alignment and the role of $\Omega_{m,\rm cosm}$

With the correct cosmological background $\Omega_{m,\rm cosm} = 0.308881$ governing the Poisson source in the growth equation, both S_8 and $f\sigma_8$ predictions align closely with ΛCDM:

- S_8 **(weak-lensing amplitude):** The IS linear-growth estimate is $S_8 = 0.8256 \pm 0.006$, 2.7σ above KiDS-1000/DES-Y3 — slightly *better* than ΛCDM (3.0σ/3.3σ);

the conservative CLASS amplitude ceiling is $S_8 = 0.827$ (2.7σ KiDS-1000). Both numbers hold A_s *fixed* to Planck, which imports the Planck–shear discrepancy; A_s is free in the model, and fitting it to the raw KiDS-1000 $\xi_\pm$ leaves $S_8 = 0.7559 \pm 0.0189$, 0.10σ (Paper 6). There is no weak-lensing tension for the single-sector IS framework to resolve.

- $f\sigma_8$ **(RSD growth rate):** SSEE predicts $f\sigma_8(z = 0.5) = 0.471$, matching the BOSS DR12 value of 0.458 ± 0.038 at $+0.35\sigma$. The mean tension across six RSD surveys is 0.70σ, essentially identical to ΛCDM (0.73σ).

The two observables are consistent: once $\Omega_{m,\mathrm{cosm}} = 0.308881$ sets the gravitational background, SSEE differs from ΛCDM only through the slightly lower matter fraction and the DE equation-of-state history $w(z)$ with $w_0 = -0.839950$, $w_a = -0.669975$. This drives a $\sim$0.3% enhancement of $\mathcal{G}$ and σ_8, which translates to a marginally different S_8.

The apparent weak-lensing tension does not survive contact with the raw data. A two-sector ϕ-DM construction was once proposed to remove it by free-streaming suppression; Paper 6 retracts that construction and shows the tension was an artefact of holding A_s at its Planck value. With A_s free, the single sector fits the 225 raw KiDS-1000 $\xi_\pm$ points at $S_8 = 0.7559 \pm 0.0189$.

7.4 The H_0 tension and SSEE

In SSEE the global rate is the output of the SH0ES cascade, $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) = 67.962 \pm 0.968$ km s^{-1} Mpc^{-1} (the measured local rate is the input; the pure number $3(\varphi + \pi)^2$ is the comparison target, not an input [Almeida Vallejo, 2026e]), consistent with Planck 2018 (67.36 ± 0.54) at 1.1σ and with DESI DR2. The IS mechanism does not modify H_0 directly, since the viscous pressure Π_{bulk} averages to zero at early times ($z > 5$) when $\Omega_{\mathrm{DE}} \to 0$. The H_0 tension between Planck and SH0ES therefore remains at the same level in SSEE as in ΛCDM — SSEE does not worsen nor resolve the H_0 tension, but introduces a new prediction for σ_8 and S_8 that partially resolves the lensing tension.

7.5 Causal structure and Hiscock-Lindblom stability

The IS theory satisfies the Hiscock-Lindblom stability criteria [Hiscock and Lindblom, 1983]: positivity of entropy production and causal propagation, provided $\tau_\Pi > 0$, $\zeta \geq 0$. We have shown that SSEE satisfies these conditions (§2.3). Moreover, the signal speed $v_{\mathrm{sig}} = c$ saturates the causality bound, meaning SSEE IS dark energy is maximally causal: perturbations propagate at the speed of light, with no superluminal modes. This distinguishes SSEE from first-order (Eckart) viscous theories, which are acausal [Hiscock and Lindblom, 1985] and would introduce spurious instabilities in the CMB power spectrum at $\ell > 1000$.

7.6 Falsifiable predictions summary

8 Conclusions

We have performed a linear Israel-Stewart causal viscous perturbation analysis of SSEE dark energy in the sub-horizon regime, addressing three physically motivated questions.

Table 7: Falsifiable predictions of SSEE IS dark energy, with observational targets and current/future datasets capable of testing each prediction. A single failure at the stated level would falsify SSEE.

Prediction	SSEE value	Test level	Dataset / Timeline
c_s^2 of the dark-energy field (Paper 7; the IS fluid limit is 0, OP-22b)	0.021284	$\lvert c_s^2\rvert < 0.05$	CMB lensing/$T_{\ell\ell'}$ cross [Calabrese et al., 2011] / current
S_8, A_s fixed to the model's own CMB fit (Paper 6)	0.8273 (no free cosmological parameter)	0.04σ from KiDS-Legacy 0.8265 ± 0.0176	KiDS-Legacy / current; Euclid / 2027
$\gamma_{\rm IS}$	0.5504 ± 0.0003	γ degenerate with ΛCDM at current precision	DESI full shape, Euclid / 2026–2028
$f\sigma_8(z = 0.5) = 0.471$	0.471	$+0.35\sigma$ BOSS; mean 0.70σ (6 surveys)	DESI full-shape / 2026
D_1 ratio $\mathcal{G}$	1.0032	0.3% enhancement; below near-term ($\gtrsim$ 1%) sensitivity	Euclid weak lensing / 2027
No DE clustering ($\lvert\delta_{\rm DE}/\delta_m\rvert < 10^{-3}$)	$< 10^{-3}$ at $k > 20\,H_0/c$	cosmic variance	Future 21cm surveys / 2030+
MIRA $= (3\varphi + \pi)/4$ (algebraic; the two-sector basis is retracted, Paper 6 — see OP-8)	1.998924	enters only the local screening fraction $f_{\rm screen}$ (Paper 9), not Ω_m	local H_0 ladders (SH0ES, CCHP) / current

Q1 — Gradient stability (proved analytically). The SSEE structure constants satisfy the exact algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\rm DE} = |w_0|$, which forces $c^2_{s,\rm eff}(k \to \infty) = w_0 + |w_0| = 0$ to machine precision. This is a construction-level identity within the SSEE constants — not a numerical coincidence or parameter tuning, but also not a theorem about arbitrary IS models. The critical wavenumber $k_{\rm crit} \simeq 0.456\,H_0/c < H_0/c$ lies outside the sub-horizon regime; all observationally relevant modes are causally stabilised. The margin between $k_{\rm crit}$ and the horizon ($\sim$ factor 2.2) is comfortable at tree level. A future one-loop treatment would test the robustness of this regime to QFT corrections; we record

this as an open extension question rather than a current limitation. Moreover, the signal speed saturates the causality bound ($v_{\rm sig} = c$), placing SSEE at the maximally causal, minimally fine-tuned configuration consistent with thermodynamic stability.

Q2 — MIRA factor (pure algebraic axiom). Linear IS perturbations yield $\mathcal{R} = 0.9897 \pm 0.0167$ at $k \geq 10\,H_0/c$, consistent with no enhancement. Its content is a bound on dark-energy clustering, $|r| \leq 0.0175$ (3.9% of the $r^* = 0.4464$ that the former target $\mathrm{MIRA}_{\rm alg} = 1.9989$ would have required): the MIRA factor is not a perturbative phenomenon. (That target was itself withdrawn in 2026-06, OP-8, so no "discrepancy" against it is claimed.) Furthermore, numerical integration of the background IS Friedmann equations yields an effective ratio of 3.04, failing to reproduce the matter factor at the background level — a figure retained as the historical record of that route closing, since the factor it targeted was itself withdrawn in 2026-06 (OP-8). We conclude that the MIRA factor does not emerge from classical Israel–Stewart fluid mechanics. A two-sector dark matter structure was once proposed (Paper 6) as its physical basis, on the grounds that two near-equal densities would produce MIRA ≈ 2 algebraically; that proposal is retracted, so MIRA has no proposed carrier at present and survives only as the algebraic invariant entering $f_{\rm screen}$ (Paper 9).

Q3 — Growth rate and S_8 (SSEE tracks ΛCDM closely). The IS growth index $\gamma_{\rm IS} = 0.5504 \pm 0.0003$ is numerically indistinguishable from $\gamma_{\Lambda\rm CDM} \simeq 0.55$. With $\Omega_{m,\rm cosm} = 0.308881$ governing the Poisson source, the growth factor ratio is $\mathcal{G} = 1.0032$, giving $\sigma_8^{\rm SSEE} = 0.8136 \pm 0.006$ and an IS linear-growth $S_8^{\rm IS} = 0.8256 \pm 0.006$ (2.7σ above KiDS-1000, slightly *better* than ΛCDM at 3.0σ). The conservative CLASS amplitude ceiling, $S_8^{\rm ceiling} = 0.827$ (2.7σ KiDS-1000), coincides with the IS linear-growth estimate above: once massive neutrinos are included, the two independent routes agree. The mean $f\sigma_8$ tension across six RSD surveys is 0.70σ, essentially identical to ΛCDM (0.73σ). There is no weak-lensing tension left for the IS sector to resolve: against the raw KiDS-Legacy $\xi_\pm$, with A_s fixed to the value the model's own CMB fit supplies, a single matter sector with no free cosmological parameter predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured; the earlier fit to KiDS-1000 with A_s free gave $S_8 = 0.7559 \pm 0.0189$, 0.10σ (Paper 6). The $S_8 = 0.8256$ quoted above is a ceiling obtained with A_s fixed to Planck, which imports the Planck–KiDS offset rather than predicting it. The ϕ-DM construction that once appeared here as the required resolution is retracted with the particle.

Together, these results establish that SSEE dark energy is: (i) causally and thermodynamically stable (Q1), (ii) predictive of a growth history essentially identical to ΛCDM once $\Omega_{m,\rm cosm} = 0.308881$ governs the growth equation (Q3), with $f\sigma_8$ tension of 0.70σ (vs. 0.73σ for ΛCDM), and (iii) strictly bounded: the MIRA factor cannot be derived from IS dynamics (Q2); the physical basis previously attributed to Paper 6's two-sector ϕ-dark matter construction is **withdrawn**: that construction and its particle are retracted (Paper 6). MIRA's algebraic value $(3\varphi + \pi)/4$ is unchanged and it survives in the model only through the screening fraction $f_{\rm screen}$ (Paper 9); **what remains open is what, if anything, provides its physical basis** — an explicitly declared gap, not a filled one. The primary falsifiable prediction in the growth sector is $S_8 = 0.8273$, obtained with A_s fixed to the value the model's own CMB fit supplies and no free cosmological parameter, against 0.8265 ± 0.0176 from the raw KiDS-Legacy $\xi_\pm$ (Paper 6); a future release placing S_8 more than 3σ from it falsifies the growth sector. The two-sector WDM correction once quoted here is retracted.

Data Availability

The Python script `src/ssee_paper5_IS_perturbations.py` (including the growth rate and S_8 computations) and all output figures are available in the SSEE repository at https://github.com/mikealmeida1721/SSEE-Cosmologia (Zenodo DOI: 10.5281/zenodo.19654118). All computations use publicly available Python packages (NumPy, SciPy, Matplotlib).

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A Derivation of the IS Dispersion Relation

We derive Eq. (22) from the linearised IS equations in Minkowski space (the cosmological curvature terms are subdominant in the sub-horizon limit).

Linearised IS system. Write $\rho = \bar{\rho} + \delta\rho$, $p = \bar{p} + \delta p$, $\Pi = \bar{\Pi} + \delta\Pi$, $u^\mu = (1, \mathbf{v})$ with $|\mathbf{v}| \ll 1$. The continuity and Euler equations give:

$$\dot{\delta\rho} + (\bar{\rho} + \bar{p})\,\nabla \cdot \mathbf{v} = 0, \tag{48}$$

$$(\bar{\rho} + \bar{p})\,\dot{\mathbf{v}} + \nabla(\delta p + \delta\Pi) = 0. \tag{49}$$

The IS relaxation equation linearises to:

$$\tau_\Pi\,\dot{\delta\Pi} + \delta\Pi = -\zeta\,\nabla \cdot \mathbf{v}. \tag{50}$$

Plane-wave ansatz. For modes $e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}$:

$$\delta\Pi = -\frac{i\zeta k}{\tau_\Pi(-i\omega) + 1}\, v_\parallel, \tag{51}$$

where $v_\parallel = \mathbf{k} \cdot \mathbf{v}/k$ is the longitudinal velocity.

Effective sound speed. Combining with the Euler equation — whose inertia is the enthalpy $\bar{\rho} + \bar{p}$, as written above, *not* $\bar{\rho}$:

$$\omega^2 = \left[c_{s,\mathrm{bare}}^2 + \frac{\zeta/[\tau_\Pi(\bar{\rho} + \bar{p})]}{1 + (\omega\tau_\Pi)^{-2}}\right] k^2. \tag{52}$$

In the IS limit $|\omega|\tau_\Pi \gg 1$ (short wavelengths):

$$\omega^2 \approx \left(c_{s,\mathrm{bare}}^2 + \frac{\zeta}{\tau_\Pi\,(\bar{\rho} + \bar{p})}\right) k^2 = c_{s,\mathrm{eff}}^2\, k^2. \tag{53}$$

Using the SSEE normalisation $\zeta = \tilde{\zeta}\,(\rho_{\mathrm{DE}} + p_{\mathrm{DE}})H_0$ (Section 2.2):

$$c_{s,\mathrm{eff}}^2 = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta} H_0}{\tau_\Pi} = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}, \tag{54}$$

recovering Eq. (22). The full k-dependent interpolation $c_{s,\mathrm{eff}}^2(k) = c_{s,\mathrm{bare}}^2 + (\tilde{\zeta}/\tau_\Pi H_0)\,x^2/(1 + x^2)$ follows from retaining the full ω dependence.

B Eigenvalue Analysis of the IS Perturbation Matrix

The quasi-static 4-variable system $\mathbf{Y} = [\delta_m, v_m, \delta_{\rm DE}, v_{\rm DE}]^\top$ has Jacobian matrix $\mathbf{J}$ (evaluated at $z = 0$, $k = 10\,H_0/c$):

$$\mathbf{J} = \begin{pmatrix} 0 & -k/H & 0 & 0 \\ -3\Omega_m/2k & -1 & -3\Omega_{\rm DE}/2k & 0 \\ 0 & 0 & 0 & -(1+w)k/H \\ -3\Omega_m/(2k(1+w)) & 0 & -c_s^2 k/((1+w)H) & -\mathcal{F} \end{pmatrix}, \tag{55}$$

where $\mathcal{F} = (1 - 3c_s^2) + \tilde{\zeta}(k/H)^2 \approx 186$ at $k = 10$, $z = 0$.

The eigenvalues λ_i (computed numerically from $\det(\mathbf{J} - \lambda\mathbf{I}) = 0$):

$$\lambda_1 \approx +0.15 \quad \text{(growing matter mode)}, \tag{56}$$

$$\lambda_2 \approx -1.15 \quad \text{(decaying matter mode)}, \tag{57}$$

$$\lambda_3 \approx +0.02 \quad \text{(IS-damped DE, slowly growing)}, \tag{58}$$

$$\lambda_4 \approx -\mathcal{F} \approx -186 \quad \text{(fast IS relaxation)}. \tag{59}$$

The large negative eigenvalue $\lambda_4 \approx -186$ drives $v_{\rm DE}$ to its quasi-static value on a timescale $\sim (186\,H)^{-1}$, justifying the quasi-static approximation. The small positive $\lambda_3 \approx +0.02$ means DE perturbations grow very slowly compared to matter ($\lambda_1 \approx 0.15$), consistent with $|\delta_{\rm DE}/\delta_m| \ll 1$ at $z = 0$. No eigenvalue has a growth rate exceeding λ_1: the system is perturbatively stable with the matter growing mode dominating.

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